

Breaking Theoretical Accuracy Ceilings of 0D Compositional Models in Double Perovskites via Interpretable Physics-Gated Routing Architecture

Highlights

- Physics-gated domain routers resolve zero-inflation in 0D double perovskite ML.
- Harrison tight-binding and tie-line engines elevate 0D models beyond SOTA limits.
- Achieved 109.6% of theoretical limit for formation energy (71.26% R^2 score).
- Robust out-of-distribution validation across 5,000 double perovskite crystals.
- Discovered interpretable analytical equations for four DFT ground-state targets.